

RADIOACTIVE DECAY

The decay of an unstable nucleus is random in time, i.e. it is not possible to predict when a particular atom will decay. This is analogous to predicting how many times you have to throw a die to get a six.

MEASUREMENTS

DECAY STATISTICS

Place a radioactive source in front of the counter window such that the count rate is about 10 events per 10 s (on average). Measure 25 times the number of events in 10 s.

Source (isotope):

Distance between source and counter:

Diameter of counter window:

Events										

For the same setup (same position of the source), take 50 measurements of the events in 5 seconds.

Events										

ANALYSIS

- Draw a histogram for the first data series. Add the data from the second series to the same histogram (with a different colour). Describe and explain the differences between the two histograms.
- Fit normal distributions to the histograms. Compare the fitted values for the mean and standard deviation with the values calculated using the measured data directly.

ROLLING DICE

Roll 24 dice (*active nuclei*). Record the number of dice showing six spots (*disintegrations*). Repeat this process 50 times.

ΔN										

ANALYSIS

- Graph a histogram showing how many times each number of six spots occurred (zero sixes, one six, two sixes, three sixes, etc.). Add the values predicted by a binomial distribution.
- Collect the data of all the other lab teams and make a histogram for the combined data. Fit a binomial distribution to the data points, using the probability and the number of dice as fit parameters. Compare the calculated values with the predicted values.

ADDITIONAL TASKS

- Simulate the decay of a radioactive substance with real or simulated dice. Start with a large number of dice and roll all of them. Remove the dice showing six spots („decayed“). The remaining dice are still „active“. Repeat the process until only a few are left. Graph the number of active dice vs the number of rolls and fit an exponential decay to the data. Compare the calculated half life (from fit parameter) with the expected one (based on the probability of a six).
- Estimate the number of active nuclei based in the radioactive source used for experiment 1. Your calculation should be based on the count rate, the half life of the isotope, the diameter of the counter window, and the distance between source and counter.
- Write a program to simulate the rolling of dice (experiment 2). Investigate how, for a large number of trials, the simulation approaches the binomial distribution.
- With the help of your program, investigate cases where the Poisson distribution or the normal distribution is a good approximation to the binomial distribution..
- For a normal distribution, 68 % of the values are within a *standard deviation* from the mean value. Determine a „visual“ standard deviation for your data sets by adjusting the range around the mean such that this rule is fulfilled. Compare the results to the calculated values (e.g. with `numpy.std`).